

Autonomous AI Agent Architectures: Reasoning, Planning, and Knowledge Representation

1. Abstract & Introduction

Modern Artificial Intelligence (AI) systems are transitioning from static input-output mappers to autonomous agents capable of reasoning, long-term planning, and environment interaction. This paper presents an integrated agent architecture combining symbolic knowledge representation with heuristic search algorithms (A^* , Monte Carlo Tree Search). We evaluate decision accuracy, goal satisfaction, and plan execution efficiency across dynamic decision environments.

2. Knowledge Representation & Planning

Knowledge representation relies on description logics and structured graph databases. The planning subsystem utilizes state-space search with admissible heuristics to formulate multi-step action sequences. Multi-agent coordination mechanisms resolve resource conflicts via automated bargaining protocols.

3. Empirical Results & Discussion

Experimental evaluations demonstrate a 34% improvement in plan execution success compared to classical rule-based systems. Memory footprint during heuristic graph expansion remains bounded within $O(b^d)$ limits.

4. Conclusion

Combining symbolic AI planning with statistical reasoning produces resilient autonomous agents suitable for high-stakes decision-making environments.